

Kinematic Restitution Balancing (KRB)

A Low-Cost Analytical Fix for Energy Gain in Velocity-Level Impulse Solvers

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Abstract

In discrete physics simulations using velocity-level impulse solvers (Sequential Impulse / Projected Gauss-Seidel), energy gain is a common numerical artifact. This paper introduces **Kinematic Restitution Balancing (KRB)**, a per-constraint correction for that leak in both unilateral constraints (collisions) and bilateral constraints (joints). By compensating for force-induced velocity drift (Component A) and auditing potential-energy shifts during position correction (Component B), KRB removes the two usual analytic energy sources inside a single-pass SI/PGS pipeline, without a second position pass. It does not claim a global energy invariant; evaluation and limitations sections state the coupled-constraint gap that follows from the per-constraint audit.

1. Introduction

Velocity-level sequential-impulse (SI) solvers [Catto 2005] are the standard real-time rigid-body approach: constraints are solved after explicit force integration with a fixed iteration count. KRB is a drop-in *preSolve* energy audit for that pipeline. It does not add solver passes, replace PGS, or switch to position-based dynamics.

Two analytic sources inject artificial mechanical energy during constraint resolution.

Force-integration drift. With symplectic Euler integration, velocity is updated before the constraint solve:

$$v_{\text{solver}} = v_t + a_{\text{ext}} \cdot \Delta t. \quad (1)$$

Restitution and joint bias see v_{solver} instead of the true relative velocity v_{impact} . For a perfect bounce ($e = 1$), the launch velocity gains $a_{\text{ext}} \cdot \Delta t$ extra speed per frame. For

a joint at rest, the solver sees gravity-induced velocity and fights it every frame at the velocity level.

Potential-energy shift from position correction. Baumgarte stabilization [Baumgarte 1972] feeds constraint error back as a velocity bias $v_{\text{bias}} = \beta C / \Delta t$, displacing bodies by Δh to resolve penetration or joint drift. That displacement increases potential energy (mgh) without reducing kinetic energy. The added PE converts to KE on later frames, causing runaway energy growth.

Split impulse and equivalent decoupled position passes [Catto 2014] keep position-correcting impulses off the physical velocity used for restitution, which removes the *immediate* bounce-from-correction artifact. They do not tax the potential-energy work of the remaining displacement Δh . Nonlinear Gauss–Seidel (NGS) position solvers likewise omit that audit [Catto 2024]. Component B closes that gap inside a single-pass SI profile. KRB is not a dual-pass temporal solver, variational integrator, or PBD/XPBD stack.

KRB applies an analytic PE/KE balance to the *expected* correction displacement Δh at constraint setup, for contacts and distance joints. The audit is per-constraint; coupled graphs can still exhibit offsets when one constraint forces work on another (see evaluation and limitations).

2. Method

KRB performs two independent corrections during constraint `preSolve`.

2.1. Component A: Force velocity compensation

Store the velocity increment from external forces during integration ($v_{\text{force}} = a_{\text{ext}} \cdot \Delta t$ per body) and compute the true relative normal velocity:

$$v_{\text{impact}} = v_{\text{relative}} - (v_{\text{force},B} - v_{\text{force},A}). \quad (2)$$

Component A applies to all constraints whenever symplectic Euler integration is used, independent of the bias method.

2.2. Component B: Kinematic energy balancing

Adjust the launch or bias velocity to account for work done by external forces over the expected correction displacement Δh :

$$v_{\text{surf}} = \sqrt{\max(0, v_{\text{impact}}^2 + 2(\mathbf{a}_{\text{ext}} \cdot \mathbf{n})\Delta h)}, \quad (3)$$

$$v_{\text{final}} = e \cdot v_{\text{surf}} = e \sqrt{\max(0, v_{\text{impact}}^2 + 2(\mathbf{a}_{\text{ext}} \cdot \mathbf{n})\Delta h)}. \quad (4)$$

For ground collisions ($\mathbf{a}_{\text{ext}} \cdot \mathbf{n} < 0$), Component B taxes launch velocity to pay for increased PE; for ceiling collisions ($\mathbf{a}_{\text{ext}} \cdot \mathbf{n} > 0$), it credits velocity for work against external forces.

2.3. Effective displacement prediction

When velocity and position phases are decoupled, the kinematic bounce velocity v_{launch} partially resolves penetration d before the position solver runs:

$$d_{\text{eff}} = \max(0, d - (v_{\text{launch}} \cdot \Delta t)). \quad (5)$$

Many engines clamp per-iteration position correction to Δh_{max} (e.g. `MAX_POSITION_CORRECTION`). The audit must match the work actually performed:

$$\Delta h = \min(d_{\text{eff}}, \Delta h_{\text{max}}) \cdot \Gamma, \quad (6)$$

where $\Gamma = 1 - (1 - \beta)^n$ for n position iterations with Baumgarte factor β .

3. Constraint application

3.1. Contacts and speculative gaps

For physical overlap ($d > 0$), both Component A and Component B are required.

Speculative contacts are created before overlap ($d < 0$, a gap) to prevent tunneling. Because no position-correction work occurs yet, $\Delta h = 0$ and Component B is omitted; Component A remains critical so gravity-induced velocity is not treated as impact speed.

Apply restitution bias only when restitution is enabled and either the corrected normal velocity exceeds a small threshold, or a speculative gap is closing faster than the gap width allows. Resting contacts must not receive an $e = 1$ launch bias.

3.2. Distance joints and the zero-velocity paradox

Distance joints target zero relative velocity along the rod. Omitting Component B entirely leaks energy when Baumgarte stretch correction does work against gravity. Component B must be applied, but capped so a resting joint is not paralyzed.

Let $v_{\text{bias}} = \beta C / \Delta t$ be the ideal correction velocity for constraint error C . The audited correction is

$$v_{\text{bias_actual}} = \sqrt{\max(0, v_{\text{bias}}^2 - 2(\mathbf{a}_{\text{ext}} \cdot \mathbf{n})(\beta C))}. \quad (7)$$

If kinetic energy cannot pay the PE tax, the joint retains a microscopic Baumgarte sag—bounded audit rather than infinite stiffness at rest. This paper’s setup recipe covers contacts and *distance* joints; other joint types are outside the listed implementation.

4. Implementation

KRB runs at contact and distance-joint `preSolve`, before the velocity iteration loop. Prerequisites: symplectic Euler with per-body `forceVelocity` ($a_{\text{ext}}\Delta t$), Baumgarte position correction with factor β , position iteration count n , and penetration `slop`.

4.1. Contact setup

```
// Component A: true impact normal velocity
forceVn = dot(bodyB.forceVelocity - bodyA.forceVelocity, n)
relativeVn = vn - forceVn

vBounce = -e * relativeVn
shouldBounce = enableRestitution &&
    (relativeVn < -RESTITUTION_THRESHOLD ||
     (depth < 0 && relativeVn < depth / dt))

if (shouldBounce) {
    if (depth > 0) {
        depthAfterVelocity = max(0, (depth - slop) - vBounce * dt)
        cumulativeFactor = 1 - pow(1 - beta, n)
        expectedDisplacement =
            min(depthAfterVelocity, MAX_POSITION_CORRECTION) * cumulativeFactor
    } else {
        expectedDisplacement = 0 // speculative: Component A only
    }
    accVn = forceVn / dt
    workTerm = 2 * accVn * expectedDisplacement
    vSurfSq = relativeVn * relativeVn + workTerm
    vFinal = e * sqrt(max(0, vSurfSq))
    bias = (depth < 0) ? -max(vFinal, depth / dt) : -vFinal
} else if (depth < 0) {
    bias = -depth / dt
} else {
    bias = 0
}
```

Listing 1. Contact `preSolve` bias (Component A + B).

4.2. Distance-joint setup

```
float forceVn = (bodyB->forceVelocity - bodyA->forceVelocity).dot(normal);
float v_bias_ideal = beta * C / dt;
float expectedDisplacement = v_bias_ideal * dt;
float accVn = forceVn / dt;
float workTerm = 2.0f * accVn * expectedDisplacement;
float v_bias_sq = v_bias_ideal * v_bias_ideal;
```

```
float adjusted_v_bias_sq = max(0.0f, v_bias_sq - workTerm);
float v_bias_actual = sqrt(adjusted_v_bias_sq);
bias = (v_bias_ideal > 0 ? v_bias_actual : -v_bias_actual) - forceVn;
```

Listing 2. Distance-joint `preSolve` bias.

During the velocity solve, use the stored bias; Component A’s `−forceVn` is folded into `bias` so `relative_vn` is not double-corrected each iteration.

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